
ANSWERS FOR THE FINAL EXAM

Algebra 1

2022 / 2023

Duration : 03 hours

Exercise 01 :

1. We have $f([-1, 1]) = [0, 1]$ and $f^{-1}([0, 1]) = [-1, 1]$, so $f^{-1}(f([-1, 1])) = [-1, 1]$. (0.5 pt)
2. We have $(1, 1)R_1(1, 2)$ and $(1, 2)R_1(1, 1)$ but $(1, 1) \neq (1, 2)$. So R_1 is not antisymmetric. (0.75 pt)
3. We have $0R_22$ but $2 \not R_20$, so R_2 is not symmetric. Also, $0R_21$ and $1R_20$ but $1 \neq 0$, so R_2 is not antisymmetric. (0.75 pt)
4. Let $a \neq 0$, b and c be elements of an integral domain such that $ab = ac$, thus $a(b - c) = ab - ac = 0$. Since $a \neq 0$, then $b - c = 0$ and so $b = c$. (0.75 pt)
5. $\mathbb{Z}$ is a subring of $\mathbb{Q}$ but $\mathbb{Z}$ is not an ideal of $\mathbb{Q}$ because $1 \in \mathbb{Z}$ and $\frac{1}{2} \in \mathbb{Q}$ but $1 \times \frac{1}{2} \notin \mathbb{Z}$. (0.75 pt)
6. Let $r \in R$. We have $\varphi(-r) + \varphi(r) = \varphi(-r + r) = \varphi(0_R) = 0_{R'}$ and so $\varphi(-r) = -\varphi(r)$. (0.5 pt)
7. $\mathbb{Z}$ is a commutative ring with unity but 2 is neither invertible nor zero-divisor. (0.75 pt)
8. Since $P \in \mathbb{R}[X]$, then $\forall z \in \mathbb{C}$ we have $P(\bar{z}) = \overline{P(z)}$ and so $P(\bar{z}_0) = \overline{P(z_0)} = 0$. (0.75 pt)
9. Since $\deg(X^4 + 1) = 4 > 2$, then $X^4 + 1$ is reducible in $\mathbb{R}[X]$. (We have $X^4 + 1 = (X^2 + \sqrt{2}X + 1)(X^2 - \sqrt{2}X + 1)$). (0.5 pt)
10. Assume that $X^3 - 2$ is reducible in $\mathbb{Q}[X]$. Since $\deg(X^3 - 2) = 3$, then it is divisible by a polynomial of degree 1 and so it has a root in $\mathbb{Q}$. But $X^3 - 2$ has no root in $\mathbb{Q}$, thus $X^3 - 2$ is irreducible in $\mathbb{Q}[X]$. (0.5 pt)

Exercise 02 :

1. Since $\deg(X^3 - X^2) = 3$, then

$$X^{n+2} + X^6 + (n-1)X^5 + 3X^4 + 2(n-1)X^3 + 11X^2 + 9(n-1)X + 9 = Q(X)(X^3 - X^2) + aX^2 + bX + c, \quad (1)$$

where $Q(X) \in \mathbb{R}[X]$ and $a, b, c \in \mathbb{R}$ (0.25 pt). Taking $X = 0$ in (??), we obtain $c = 9$. Differentiating both sides of (??) with respect to X , then taking $X = 0$ we get $b = 9(n-1)$. Finally, by taking $X = 1$ in (??) we obtain $a = 3n + 13$. Therefore, the remainder is $(3n + 13)X^2 + (9n - 9)X + 9$. (0.75 pt)

2. We have

$$P_0(X) = X^6 - X^5 + 3X^4 - 2X^3 + 11X^2 - 9X + 9 = (X^2 - 1)(X^4 - X^3 + 4X^2 - 3X + 15) - 12X + 24,$$

$$X^2 - 1 = (-12X + 24) \left(-\frac{X}{12} - \frac{1}{6} \right) + 3 \text{ and } -12X + 24 = 3(-4X + 8).$$

So $\gcd(P_0(X), X^2 - 1) = 1$ (0.5 pt). Furthermore using the above formula, we get

$$-12X + 24 = P_0(X) - (X^2 - 1)(X^4 - X^3 + 4X^2 - 3X + 15) \text{ and } 3 = X^2 - 1 - (-12X + 24) \left(-\frac{X}{12} - \frac{1}{6} \right).$$

Thus,

$$U(X)P_0(X) + V(X)(X^2 - 1) = 1,$$

where

$$U(X) = \frac{X}{36} + \frac{1}{18} \quad \text{and} \quad V(X) = \left(-\frac{X}{36} - \frac{1}{18}\right)(X^4 - X^3 + 4X^2 - 3X + 15) + \frac{1}{3}. \text{(1.5 pt)}$$

3.(a) $P_n(i) = 0$ iff $i^6 + (n-1)i^5 + 3i^4 + 2(n-1)i^3 + 11i^2 + 9(n-1)i + 9 = 0$ iff $-1 + (n-1)i + 3 + 2(n-1)(-i) - 11 + 9(n-1)i + 9 = 0$ iff $(8n-8)i = 0$ iff $n = 1$. (1 pt)

(b) Since i is a root of $P_1(X)$ and $P_1(X) \in \mathbb{R}[X]$, so $-i$ is also a root of $P_1(X)$, thus $X^2 + 1$ divides $P_1(X)$ (0.5 pt). We have

$$P_1(X) = (X^2 + 1)(X^4 + 2X^2 + 9) = (X^2 + 1)((X^2 + 3)^2 - (2X)^2) = (X^2 + 1)(X^2 - 2X + 3)(X^2 + 2X + 3). \text{(0.5pt)}$$

All the factors of the above decomposition are with degree 2 and negative discriminant and so all the factors of the above decomposition are irreducible in $\mathbb{R}[X]$. On the other hand, the decomposition of $P_1(X)$ in $\mathbb{C}[X]$ is

$$P_1(X) = (X - i)(X + i)(X - 1 - \sqrt{2}i)(X - 1 + \sqrt{2}i)(X + 1 - \sqrt{2}i)(X + 1 + \sqrt{2}i) \text{ (0.25 pt)}.$$

(c) We have

$$\frac{2(X-1)^2(X^2+2X+3)}{P_1(X)} = \frac{2(X-1)^2}{(X^2+1)(X^2-2X+3)} = \frac{AX+B}{X^2+1} + \frac{CX+D}{X^2-2X+3}, \quad (2)$$

where $A, B, C, D \in \mathbb{R}$. Multiplying both sides by $X^2 + 1$ and taking $X = i$, we obtain

$$Ai + B = \frac{2(i-1)^2}{i^2 - 2i + 3} = \frac{2(i-1)^2}{2 - 2i} = 1 - i,$$

so $A = -1$ and $B = 1$. Taking $X = 0$ in (??), we obtain $D = 2 - 3B = -1$. Multiplying both sides of (??) by X and taking the limit when $X \rightarrow +\infty$, we obtain $C = -A = 1$. Therefore, it follows that

$$\frac{2(X-1)^2(X^2+2X+3)}{P_1(X)} = \frac{-X+1}{X^2+1} + \frac{X-1}{X^2-2X+3} \text{(1.5pt)}.$$

Exercise 03 :

1. Let p be a prime and G be a group such that $|G| = p$. Then G contains more than one element. Let $g \in G$ such that $g \neq e_G$. Then $\langle g \rangle$ contains more than one element. Since $\langle g \rangle$ is a subgroup of G , by Lagrange's theorem, $|\langle g \rangle|$ divides p . Since $|\langle g \rangle| > 1$, $|\langle g \rangle| = p = |G|$. Hence, $\langle g \rangle = G$. It follows that G is cyclic (0.5 pt).

2. (a) Let $\bar{x}, \bar{y} \in \mathbb{Z}/n\mathbb{Z}$. We have $\varphi_k(\bar{x} + \bar{y}) = \varphi_k(\overline{x+y}) = \overline{k(x+y)} = \overline{kx+ky} = \overline{kx} + \overline{ky} = \varphi_k(\bar{x}) + \varphi_k(\bar{y})$. Then φ_k is a group endomorphism (0.5 pt).

(b) Suppose that $\varphi = \varphi_k$ for some $1 \leq k \leq n$. By the previous question φ is a group endomorphism. Conversely, if φ is a group endomorphism then for any $\bar{x} \in \mathbb{Z}/n\mathbb{Z}$, $\varphi(\bar{x}) = \varphi(\underbrace{\bar{1} + \bar{1} + \dots + \bar{1}}_{x \text{ times}}) = x\varphi(\bar{1}) = \overline{kx}$, where $\bar{k} = \varphi(\bar{1})$, with $1 \leq k \leq n$ (0.5 pt).

(c) φ_k is a ring endomorphism iff $\forall \bar{x}, \bar{y} \in \mathbb{Z}/n\mathbb{Z}$, $\varphi_k(\bar{xy}) = \varphi_k(\bar{x})\varphi_k(\bar{y})$ iff $\forall \bar{x}, \bar{y} \in \mathbb{Z}/n\mathbb{Z}$, $\overline{kxy} = \overline{k^2xy}$ iff $k^2 = k \pmod{n}$ iff n divides $k(k-1)$ (0.5 pt).

(d)+(e) Using the previous result, the endomorphisms of $\mathbb{Z}/4\mathbb{Z}$ are φ_1, φ_4 , and the endomorphisms of $\mathbb{Z}/6\mathbb{Z}$ are

$\varphi_1, \varphi_3, \varphi_4, \varphi_6$ (0.5+0.5 pt).

3.(a) We have $|G| = |\mathbb{Z}/2\mathbb{Z}| \times |\mathbb{Z}/2\mathbb{Z}| = 2 \times 2 = 4$ (0.25 pt) and $G = \{(\bar{0}, \bar{0}), (\bar{0}, \bar{1}), (\bar{1}, \bar{0}), (\bar{1}, \bar{1})\}$ (0.5 pt). The addition table of G is the following (0.75 pt)

$\oplus$	$(\bar{0}, \bar{0})$	$(\bar{0}, \bar{1})$	$(\bar{1}, \bar{0})$	$(\bar{1}, \bar{1})$
$(\bar{0}, \bar{0})$	$(\bar{0}, \bar{0})$	$(\bar{0}, \bar{1})$	$(\bar{1}, \bar{0})$	$(\bar{1}, \bar{1})$
$(\bar{0}, \bar{1})$	$(\bar{0}, \bar{1})$	$(\bar{0}, \bar{0})$	$(\bar{1}, \bar{1})$	$(\bar{1}, \bar{0})$
$(\bar{1}, \bar{0})$	$(\bar{1}, \bar{0})$	$(\bar{1}, \bar{1})$	$(\bar{0}, \bar{0})$	$(\bar{0}, \bar{1})$
$(\bar{1}, \bar{1})$	$(\bar{1}, \bar{1})$	$(\bar{1}, \bar{0})$	$(\bar{0}, \bar{1})$	$(\bar{0}, \bar{0})$

(b) By Lagrange's theorem, the order of a subgroup must divide the order of the group, then the proper subgroups of G are of order 2. The subgroups of order 2 are cyclic (thanks to question 1). Then we conclude that all the proper subgroups of G are cyclic (0.5 pt).

Note that in many books, If G is a group of neutral element e , the trivial subgroup $\{e\}$ is considered as a proper subgroup.

(c) The proper subgroups of G are $F_1 = \langle (\bar{0}, \bar{1}) \rangle = \{(\bar{0}, \bar{0}), (\bar{0}, \bar{1})\}$, $F_2 = \langle (\bar{1}, \bar{0}) \rangle = \{(\bar{0}, \bar{0}), (\bar{1}, \bar{0})\}$ and $F_3 = \langle (\bar{1}, \bar{1}) \rangle = \{(\bar{0}, \bar{0}), (\bar{1}, \bar{1})\}$ (0.75 pt).

(d) Suppose ψ is a group homomorphism from $(\mathbb{Z}/4\mathbb{Z}, +)$ to $(G, \oplus)$, then $\psi(\bar{0}) = (\bar{0}, \bar{0})$ and $\psi(\bar{2}) = \psi(\bar{1}) + \psi(\bar{1}) = 2\psi(\bar{1}) = (\bar{0}, \bar{0}) = \psi(\bar{0})$ (since any element different from $(\bar{0}, \bar{0})$ is of order 2 in G) therefore ψ cannot be injective. We conclude then there is no group isomorphism between $(\mathbb{Z}/4\mathbb{Z}, +)$ and $(G, \oplus)$ (0.25 pt).

(e) When H is a group such that all its proper subgroups are cyclic we cannot say that H is cyclic. The group G is an example of non cyclic group whose proper subgroups are cyclic (0.5 pt).

4. (a) The multiplication table of G is the following (0.75 pt)

$\otimes$	$(\bar{0}, \bar{0})$	$(\bar{0}, \bar{1})$	$(\bar{1}, \bar{0})$	$(\bar{1}, \bar{1})$
$(\bar{0}, \bar{0})$	$(\bar{0}, \bar{0})$	$(\bar{0}, \bar{0})$	$(\bar{0}, \bar{0})$	$(\bar{0}, \bar{0})$
$(\bar{0}, \bar{1})$	$(\bar{0}, \bar{0})$	$(\bar{0}, \bar{1})$	$(\bar{0}, \bar{0})$	$(\bar{0}, \bar{1})$
$(\bar{1}, \bar{0})$	$(\bar{0}, \bar{0})$	$(\bar{0}, \bar{0})$	$(\bar{1}, \bar{0})$	$(\bar{1}, \bar{0})$
$(\bar{1}, \bar{1})$	$(\bar{0}, \bar{0})$	$(\bar{0}, \bar{1})$	$(\bar{1}, \bar{0})$	$(\bar{1}, \bar{1})$

(b) We have $D = \{(\bar{1}, \bar{0}), (\bar{0}, \bar{1})\}$, $U = \{(\bar{1}, \bar{1})\}$, $N = \{(\bar{0}, \bar{0})\}$ and $I = \{(\bar{0}, \bar{0}), (\bar{0}, \bar{1})(\bar{1}, \bar{0}), (\bar{1}, \bar{1})\}$ (0.25+0.25+0.25+0.25 pt).

(c) The characteristic of G is 2, since G is not the zero ring and $(\bar{1}, \bar{0}) + (\bar{1}, \bar{0}) = (\bar{0}, \bar{0})$, $(\bar{0}, \bar{1}) + (\bar{0}, \bar{1}) = (\bar{0}, \bar{0})$, $(\bar{1}, \bar{1}) + (\bar{1}, \bar{1}) = (\bar{0}, \bar{0})$ (0.25 pt).

(d) $(G, \oplus, \odot)$ is not a field since $(\bar{0}, \bar{1})$ is not invertible (0.25 pt).

5. (a) The order of $(\bar{1}, \hat{1})$ in $(\mathbb{Z}/n\mathbb{Z} \times \mathbb{Z}/m\mathbb{Z}, \boxplus)$ is the least $k \in \mathbb{N}^*$ such that $k(\bar{1}, \hat{1}) = (\bar{0}, \hat{0})$. If $k \in \mathbb{N}^*$ with $k(\bar{1}, \hat{1}) = (\bar{0}, \hat{0})$ then $(\bar{k}, \hat{k}) = (\bar{0}, \hat{0})$ then n divides k and m divides k , hence k is a common multiple of n and

m . Since $\text{lcm}(n, m)$ is the least common multiple of n and m , then $\text{lcm}(n, m)$ is the least $k \in \mathbb{N}^*$ satisfying $k(\bar{1}, \hat{1}) = (\bar{0}, \hat{0})$. Then the order of $(\bar{1}, \hat{1})$ is $\text{lcm}(n, m)$ (1 pt).

(b) The order of the group $(\mathbb{Z}/n\mathbb{Z} \times \mathbb{Z}/m\mathbb{Z}, \boxplus)$ is nm . Suppose $\text{gcd}(n, m) = 1$. Then $\text{lcm}(n, m) = nm$. Thanks to the previous question, the element $(\bar{1}, \hat{1})$ is of order nm which is the order of the group $(\mathbb{Z}/n\mathbb{Z} \times \mathbb{Z}/m\mathbb{Z}, \boxplus)$. Then the group $(\mathbb{Z}/n\mathbb{Z} \times \mathbb{Z}/m\mathbb{Z}, \boxplus)$ is cyclic and $(\bar{1}, \hat{1})$ is a generator. Knowing that any two cyclic groups of the same order are isomorphic then $(\mathbb{Z}/n\mathbb{Z} \times \mathbb{Z}/m\mathbb{Z}, \boxplus)$ is isomorphic to $(\mathbb{Z}/nm\mathbb{Z}, +)$. Conversely, let $l := \text{lcm}(n, m)$. Suppose that $d = \text{gcd}(n, m) \geq 2$, and then there exist two natural numbers m' and n' such that $m = dm'$ and $n = dn'$ with $\text{gcd}(n', m') = 1$. We know that $\text{lcm}(n, m) \times \text{gcd}(n, m) = mn$. Hence, $l = dm'n' = m'n = mn' < mn$. Let $(\bar{x}, \hat{y}) \in \mathbb{Z}/n\mathbb{Z} \times \mathbb{Z}/m\mathbb{Z}$, we have $l(\bar{x}, \hat{y}) = \underbrace{(\bar{x}, \hat{y}) \boxplus (\bar{x}, \hat{y}) \boxplus \cdots \boxplus (\bar{x}, \hat{y})}_{l \text{ times}} = (l\bar{x}, l\hat{y}) = (\bar{0}, \hat{0})$. Thus, any element of $\mathbb{Z}/n\mathbb{Z} \times \mathbb{Z}/m\mathbb{Z}$ has an order strictly less than mn . This contradicts the fact that $(\mathbb{Z}/n\mathbb{Z} \times \mathbb{Z}/m\mathbb{Z}, \boxplus)$ is cyclic of order mn . Therefore $\text{gcd}(n, m) = 1$ (1 pt).